

Cutting Plane Methods

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1 GLS Oracle Model

1.1 Definitions

Definition 1 (Oracle Model). *The following are algorithmic problems for closed convex set $K \subseteq \mathbb{R}^n$:*

MEMbership($x \in \mathbb{R}^n, K \subseteq \mathbb{R}^n$):

Output: YES if $x \in K$; NO otherwise.

*SEP*aration($x \in \mathbb{R}^n, K \subseteq \mathbb{R}^n$):

Output: YES if $x \in K$;

otherwise output separating hyperplane w such that $\langle w, x \rangle > \sup_{y \in K} \langle w, y \rangle$;

*OPT*imization($w \in \mathbb{R}^n, K \subseteq \mathbb{R}^n$)

Output: $h_K(w) = \sup_{x \in K} \langle w, x \rangle$

and optimizer $x_ \in K$ such that $h_K(w) = \langle w, x_* \rangle$;*

*VAL*idity($w \in \mathbb{R}^n, b \in \mathbb{R}, K \subseteq \mathbb{R}^n$):

Output: YES if $\langle w, x \rangle \leq b$ for all $x \in K$; NO otherwise.

*VIOL*ation($w \in \mathbb{R}^n, b \in \mathbb{R}, K \subseteq \mathbb{R}^n$):

Output: YES if $\langle w, x \rangle \leq b$ for all $x \in K$;

otherwise output $x \in K$ such that $\langle w, x \rangle > b$.

We can also consider *approximate* versions of all these oracles. For the formal definitions, see [GLS 2.1]. Many of the algorithms we study in this class can be thought of as reductions between these algorithmic problems.

1.2 Relations between Oracles

The various oracles for K and its polar dual K° are related.

Proposition 2. *In the following, we assume $B(0, \varepsilon) \subseteq \text{int}(K)$ and $B(0, \varepsilon) \subseteq \text{int}(K^\circ)$.*

- *OPTimization oracle for K can be (approximately) implemented using VIOLation or VALidity oracle for K ;*
- *MEMbership oracle for K is equivalent to VALidity oracle for K° ; and vice-versa $\text{MEM}(K^\circ)$ is equivalent to $\text{VAL}(K)$;*

- *SEP*aration oracle for K° can be implemented using a *VIOL*ation oracle for K ; $\text{VIOL}(K)$ can be implemented using $\text{SEP}(K^\circ)$ and one of $\text{MEM}(K)$ or $\text{VAL}(K^\circ)$.

Proof:

- In the following we use VAL if we only need the optimum value, and VIOL if the optimizer is also required. Using the VAL/VIOL oracles we can check if $\exists x \in K : \langle w, x \rangle \geq b$. Therefore, to compute $h_K(w) = \sup_{x \in K} \langle w, x \rangle$ approximately, we perform binary search over b .
- We first implement $\text{MEM}(K^\circ)$ using $\text{VAL}(K)$. Given input w we want to check if $w \in K^\circ$ iff $\sup_{x \in K} \langle w, x \rangle \leq 1$. For this we return the output of $\text{VAL}(w, 1, K)$.

Conversely, we want to implement $\text{VAL}(K)$ using $\text{MEM}(K^\circ)$. So given an input (w, b) we want to check if $\forall x \in K : \langle w, x \rangle \leq b$. First we use that $B(0, \varepsilon) \subseteq K$ so if $b \leq 0$ we output NO. Otherwise note that $v \in K^\circ$ iff $\sup_{x \in K} \langle v, x \rangle \leq 1$. So we can return the output of $\text{MEM}(w/b, K^\circ)$.

The dual statement, swapping K and K° , follows due to bi-duality $(K^\circ)^\circ = K$.

- We first implement $\text{SEP}(K^\circ)$ using $\text{VIOL}(K)$. Given input w we want to check if $w \in K^\circ$, and if not to output a separating hyperplane. Note $w \in K^\circ$ iff $\sup_{x \in K} \langle w, x \rangle \leq 1$, so we query $\text{VIOL}(w, 1, K)$; if the output is YES, we return YES; if the output is NO then the oracle returns $x \in K$ such that $\langle w, x \rangle > 1 \geq \sup_{v \in K^\circ} \langle v, x \rangle$, where the last step is by definition of K° , so we can output x as our separating hyperplane for K° .

Conversely, we want to implement $\text{VIOL}(K)$ using $\text{SEP}(K^\circ)$. So given an input (w, b) we want to check if $\sup_{x \in K} \langle w, x \rangle \leq b$, and if not to output $x \in K$ such that $\langle w, x \rangle > b$. First we use that $B(0, \varepsilon) \subseteq K$ so if $b \leq 0$ we output $0 \in K$ as $\langle w, 0 \rangle = 0 > b$. Otherwise we query $\text{SEP}(w/b, K^\circ)$; if the output is YES then $\sup_{x \in K} \langle w/b, x \rangle \leq 1$ so we return YES; if the output is NO then the oracle returns separating hyperplane $x \in \mathbb{R}^n$ such that $\langle w/b, x \rangle > \sup_{v \in K^\circ} \langle v, x \rangle$. If we knew this right hand side $h_{K^\circ}(x) = \sup_{v \in K^\circ} \langle v, x \rangle$ then we could output $y := x/h_{K^\circ}(x) \in K^{\circ\circ} = K$ with $\langle w, y \rangle > b$ as our violating point. But we can approximately compute this value using binary search with the VAL idity oracle for K° as shown in the first part. Equivalently, since $0 \in \text{int}(K)$, there is a point $y \in [0, x] \cap \partial K$ at the intersection of the line $[0, x]$ with the boundary of K , and this also gives a violating point $\langle w, y \rangle > b$ ($y \in \partial K$ implies $\sup_{v \in K^\circ} \langle v, y \rangle = 1$). And this boundary point we can approximately compute using binary search with the MEM bership oracle for K .

□

1.3 Function Oracles

What is the relation to function oracles? In particular, recall that we prove duality of convex functions by reducing to duality of the epigraph, which is a convex set.

Definition 3 (Fenchel dual). *For convex $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the Fenchel dual is*

$$f^*(w) := \sup_y \langle w, y \rangle - f(y).$$

Lemma 4. *For differentiable closed convex f , the supporting hyperplane for the epigraph of f at $(x, f(x))$ is given by slope $(\nabla f(x), -1)$ and value $f^*(\nabla f(x)) = \sup_y \langle \nabla f(x), y \rangle - f(y)$.*

Similarly, the supporting hyperplane for the sub-level set $L_{f(x)} := \{y \in \mathbb{R}^n \mid f(y) \leq f(x)\}$ is given by slope $\nabla f(x)$ and value $\langle \nabla f(x), x \rangle$.

Both these statements can be generalized to non-differentiable f using subgradients.

Definition 5 (Oracle Model). *The following are algorithmic problems for closed convex function $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$:*

EVALuation ($x \in \mathbb{R}^n, f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$):
Output: $f(x)$.

GRADient ($x \in \mathbb{R}^n, f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$):
Output: $(f(x), g)$ for $g \in \partial f(x)$.

Proposition 6. *In the following, we assume $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ is a closed convex function.*

- *EVALuation oracle for f can be implemented using MEMbership oracle for $\text{epi}(f)$;*
- *GRADient oracle for f can be implemented using SEParation oracle for $\text{epi}(f)$;*
- *OPTimization oracle for $\text{epi}(f)$ is equivalent to EVALuation for the Fenchel dual $f^*(w) := \sup_y \langle w, y \rangle - f(y)$.*

2 Cutting Plane Methods

Our eventual goal is to find an algorithm to solve convex programs $\min_{x \in K} f(x)$. Cutting plane methods solve this using just SEParation oracle for K and EVAL and GRAD oracles for f . We first show how to solve the much simpler FEASibility problem: given SEParation oracle for K , either find $x \in K$ or output that K is ‘small’. Note that linear optimization over K can be reduced to this: test FEASibility for

$$K' := K \cap \{x \mid \langle w, x \rangle \geq b\}.$$

and run binary search on b . Note that SEP for K' can be easily implemented using SEP for K and testing the linear inequality $\langle w, x \rangle \geq b$.

The intuition for cutting plane methods comes from binary search: say we are attempting to find some point $x \in [0, 1]^n$, and we have oracle access to coordinate queries $y_i \geq x_i$ or $y_i \leq x_i$. Then in each iteration, we should query the center of the remaining grid and eliminate the elements with chosen coordinate too large or small, depending on the output of the oracle. This procedure is optimal in the worst case, as it eliminates the maximum possible options in each iteration. This is the correct intuition for the Center of Gravity method, described below, which generalizes the procedure to arbitrary convex $K \subseteq \mathbb{R}^n$. In general, it is not clear how to query the ‘center’ of a convex body, so the next algorithm will maintain an Ellipsoid in each iteration to contain the feasible set, and update according to the separation oracle.

3 Center-of-Gravity Method

We first describe the algorithm to solve feasibility. In each iteration, we maintain a feasible convex set $K_t \supseteq K$; we query the separation oracle with the Center of Mass $c_t := \mathbb{E}_{x \sim K_t}[x] = \int_{x \in K_t} x dx$; if $c_t \in K$ then we are done; otherwise we get $\langle w, c_t \rangle > \sup_{x \in K} \langle w, x \rangle$; therefore, we know K is contained in the following halfspace

$$H_t := \{x \mid \langle w, x \rangle \leq \langle w, c_t \rangle\},$$

so we update accordingly $K_{t+1} := K_t \cap H_t$. We claim that this solves feasibility:

Theorem 7. *Given $K \subseteq B(0, R) \subseteq \mathbb{R}^n$ via separation oracle, the center-of-gravity method requires $O(n \log(R/\varepsilon))$ iterations to either*

1. *find $x \in K$;*
2. *certify K does not contain a ball of radius ε , or certify $\text{vol}(K) \leq \varepsilon^n$.*

The key step in the analysis is the following beautiful result from convex geometry:

Theorem 8 (Grunbaum's Theorem). *For convex $K \subseteq \mathbb{R}^n$, let $c := \mathbb{E}_{x \in K} x$ be the center-of-gravity of K . Then for any hyperplane $H \ni c$, the two halfspaces H_+, H_- satisfy*

$$\max\{\text{vol}(K \cap H_+), \text{vol}(K \cap H_-)\} \leq \text{vol}(K)(1 - 1/e).$$

We do not prove this in this class but refer to the excellent survey of Keith Ball: An Elementary Introduction to Modern Convex Geometry. With this result, the analysis is straightforward.

Proof: [Proof of theorem 7] If in any iteration we find $c_t \in K$ then we are done. Otherwise, in each iteration we reduce the volume by the factor stated above. Therefore in $T = O(n \log(R/\varepsilon))$ iterations we have

$$\text{vol}(K_T) \leq \text{vol}(K_0)(1 - 1/e)^T \leq \text{vol}(B(0, R)) \exp(-n \log(R/\varepsilon)) \leq \text{vol}(B(0, \varepsilon)),$$

where we used that $K \subseteq B(0, R)$. □

In the following section, we show that this is in fact the optimal query complexity possible for an algorithm using just a SEParation oracle. Of course, as stated it is not at all clear how to compute the center-of-gravity of K_t , and it turns out this is at least as hard as optimizing over K_t . Therefore in the following sections we will study the Ellipsoid algorithm, which requires more oracle queries but can be efficiently updated.

We next describe a very similar algorithm to solve general convex programs. In each iteration, we still maintain a feasible convex set K_t and query the center c_t ; if $c_t \notin K$ then we update $K_{t+1} = K_t \cap H_t$ just as in the previous algorithm; if $c_t \in K$ then we query the EVALuation and (sub)-GRADient oracle for f , to compute $(f(c_t), g)$; Finally we update $K_{t+1} = K_t \cap H_g$ where

$$H_g := \{x \mid \langle g, x - c_t \rangle \leq 0\}$$

i.e. the halfspace in the negative (sub-)gradient direction.

Theorem 9. *Given $K \subseteq B(0, R) \subseteq \mathbb{R}^n$ via SEParation oracle, and convex $f : \mathbb{R}^n \rightarrow \mathbb{R}$ via EVALuation and GRADient oracle, such that $\max_{x \in K} f(x) - \min_{x \in K} f(x) \leq F$; The center-of-gravity method requires $O(n \log(RF/\varepsilon))$ iterations to either*

1. *find $x \in K$ such that $f(x) \leq \min_{y \in K} f(y) + \varepsilon$;*
2. *certify $\text{vol}(K) \leq \varepsilon^n$.*

The proof rests on the following claim, showing the negative gradient gives a good update.

Claim 10. *For closed convex $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $x \in \text{dom}(f)$, any subgradient $g \in \partial f(x)$ gives a supporting hyperplane for the sub-level set*

$$L_{f(x)} := \{y \mid f(y) \leq f(x)\}.$$

We leave the proof as an exercise. But intuitively the negative subgradient halfspace always contains the optimizer $x_* := \arg \min_{x \in K} f(x)$, so our update maintains a body containing x_* . With this in hand, the analysis follows:

Proof: [Proof of theorem 9] We still manage to decrease the volume by a constant factor in each iteration, due to Grunbaum's theorem. And the claim above shows that $K_t \ni x_*$ for all t . If in every iteration $c_t \notin K$, then by the previous feasibility analysis we can certify small volume in $T = O(n \log(R/\varepsilon))$ iterations. Otherwise, we show that when the volume gets small enough, one of the queried center points must have small function value.

For this we analyze the follows approximate optimal set:

$$X_\delta := (1 - \delta)x_* + \delta K = \{(1 - \delta)x_* + \delta x \mid x \in K\}.$$

By convexity $X_\delta \subseteq K$, and we can bound the function on X_δ as

$$f((1 - \delta)x_* + \delta x) \leq (1 - \delta)f(x_*) + \delta f(x) \leq (1 - \delta)f(x_*) + \delta(f(x_*) + F) \leq f(x_*) + \delta F,$$

where we used convexity of f , and our assumed bound $\max_{x \in K} f(x) - \min_x f(x) \leq F$.

We also have $\text{vol}(X_\delta) = \text{vol}(\delta K) = \delta^n \text{vol}(K)$. Therefore if we reach T such that $\text{vol}(K_T) < \text{vol}(X_\delta) = \delta^n \text{vol}(K)$ then we must have some point $x \in X_\delta$ that is not contained in our current feasible set $x \notin K_T$. Let t be the first iteration where $x \in K_t, x \notin K_{t+1}$, and let c_t be the queried point with sub-gradient $g \in \partial f(c_t)$. Then we have

$$f(x) \geq f(c_t) + \langle g, x - c_t \rangle > f(c_t),$$

where the first step is by convexity, and for the second we used that $x \notin K_{t+1}$ so $\langle g, x \rangle > \langle g, c_t \rangle$. Rearranging gives

$$\min_{t \in [T]} f(c_t) \leq f(x) \leq f(x_*) + \delta F,$$

where the last step is by our approximate bound on X_δ shown above. $\square$

We will show in the following that this achieves the optimal *query* complexity with respect to the separation oracle. Of course we have not specified how to actually find the center of gravity of a given polytope. It turns out that this task is very computationally expensive (about as difficult as solving the optimization problem itself). Another variant that is more efficient uses the maximum inscribed ellipsoid of the current feasible region and queries its center. Computing the maximum inscribed ellipsoid is also difficult, but we maintain optimal query complexity with this algorithm. The analysis shows this improves the volume of the *max inscribed ellipsoid* by a constant factor in each iteration (not necessarily the volume of the feasible set). In the remainder we will discuss the Ellipsoid method: this does not keep track of all previous separating halfspaces, but instead maintains and updates a single ellipsoid that contains the feasible region. This method gives up on optimal query complexity, but each iteration can be implemented efficiently.

4 Lower Bound

Theorem 11. *Given $K \subseteq B(0, R) \subseteq \mathbb{R}^n$ via SEPARATION oracle, any algorithm solving the FEASIBILITY problem requires $\Omega(n \log(R/\varepsilon))$ oracle calls to the SEP oracle.*

Proof: [Proof Sketch] Given any convex body C , any $x \in K$, and any direction w , one of the halfspaces $H_\pm := \{y \mid \text{sign}(\langle w, y - x \rangle) = \pm\}$ must maintain at least $1/2$ volume, i.e.

$$\max\{\text{vol}(C \cap H_+), \text{vol}(C \cap H_-)\} \geq \frac{\text{vol}(C \cap H_+) + \text{vol}(C \cap H_-)}{2} = \frac{\text{vol}(C)}{2}.$$

So deciding whether K is empty or not requires the stated number of queries, as an adversary can always produce a separating hyperplane maintaining $\geq 1/2$ volume in each iteration. $\square$

5 Ellipsoid Method

Note that while the Center-of-Gravity method provably achieves optimal oracle complexity, as shown by the lower bound above, it is not clear how to implement the procedure in general: computing the centerpoint of a general convex body is at least as hard as optimizing over that body! And maintaining the sequence of intersections of halfspaces could become costly. Therefore, in this section we consider the Ellipsoid method:

Theorem 12. *Given $K \subseteq B(0, R) \subseteq \mathbb{R}^n$ via separation oracle, the Ellipsoid method requires $O(n^2 \log(R/\varepsilon))$ iterations to either*

1. *find $x \in K$;*
2. *certify K does not contain a ball of radius ε .*

The strategy involves approximating our feasible set by an Ellipsoid. This can be maintained and updated efficiently. The key geometric lemma that allows us to make progress is as follows:

Lemma 13 (Lemma 2.3 in (Bubeck)). *Let $\mathcal{E} := \{x \in \mathbb{R}^n \mid \langle (x - c), Q(x - c) \rangle \leq 1\}$ be an Ellipsoid with $Q \succ 0$; and let $H := \{x \in \mathbb{R}^n \mid \langle w, x \rangle \leq \langle w, c \rangle\}$ be a halfspace through the center of $\mathcal{E}$. Then there exists ellipsoid $\bar{\mathcal{E}}$ such that*

$$\bar{\mathcal{E}} \supset \mathcal{E} \cap H \quad \text{and} \quad \text{vol}(\bar{\mathcal{E}}) \leq \text{vol}(\mathcal{E}) \exp(-1/2n).$$

Further, this update can be efficiently computed in terms of (c, Q) as

$$\bar{c} := c - \frac{1}{n+1} \frac{Q^{-1}w}{\sqrt{\langle w, Q^{-1}w \rangle}}, \quad \bar{Q} := \left(1 - \frac{1}{n^2}\right)Q + \frac{2(n+1)}{n^2} \frac{ww^T}{\langle w, Q^{-1}w \rangle}.$$

In the following note we prove the the geometric result using a convex program. Given this result, the complexity follows from the same proof as the Center-of-Gravity algorithm, except with n factor worse oracle complexity.

Remark 14. *Why do we choose an ellipsoid as our feasible set to update? One reason is that it can be maintained by storing (Q, c) ; this is because an ellipsoid can be written as the sub-level set of a convex quadratic function. Another reason the symmetry with respect to hyperplanes: note that any halfspace that intersects the center contains exactly half the volume. Of course we do not want to explicitly maintain this intersection, but the above lemma shows that we can choose another smaller ellipsoid that gives sufficient improvement in the volume.*

Why not other geometric objects? For our grid search illustration we just maintained an axis-aligned rectangular prism, which requires storing a vertex and n lengths, one for each dimension. But it can be shown that there is a halfspace through the center of an axis-aligned prism for which there is no smaller axis-aligned prism containing the intersection! (Try to draw this example). I'm not sure if this remains true if we consider arbitrary prisms.

Another option is to maintain a simplex, i.e. the convex hull of $n+1$ points. It turns out that this gives another valid cutting plane method with similar oracle complexity.